

VANSH JAIN

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SUMMARY

Final-year B.Tech CSE (AI/ML) student building AI-powered applications end to end — GenAI/LLM integration, RAG, and AI inference APIs through backend engineering and production deployment. Ships AI systems as usable services with FastAPI/Flask and REST APIs, backed by classical ML (Scikit-learn, XGBoost) where a task calls for it, and validates everything with automated testing (63+ tests across projects). Focused on taking AI systems from experimentation to deployed, testable applications rather than notebooks alone.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL

GenAI/LLM: Generative AI, LLMs, LLM APIs, RAG, Prompt Engineering, Embeddings, Vector Databases (Chroma), Conversational Memory, AI Application Development

AI/API Deployment: FastAPI, Flask, REST APIs, API Development, Model Serving, AI Inference APIs, OAuth 2.0, Input Validation, Error Handling

Machine Learning: Scikit-learn, XGBoost, Feature Engineering, Time-Series Analysis, NLP, TF-IDF

Data/Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly, EDA, Statistical Analysis

Engineering/Deployment: Git, GitHub, Linux, Pytest, Unit & Integration Testing, Postman, CI/CD, Logging

EDUCATION

VIT Bhopal University — B.Tech, Computer Science Engineering (AI/ML)

2023–2027

CGPA: 8.04 / 10

PROJECTS

QuantEdge — AI-Powered Stock Ranking & Trading Terminal

2026

Python, AI/ML Pipeline, Kite Connect API, Streamlit, Pytest

- Built an AI-powered stock ranking system covering the full data-to-deployment workflow — ingestion → feature engineering → model training → ranking → backtesting → live data → dashboard — across ~1M OHLCV rows and 500 Nifty 500 stocks.
- Designed an 8-layer modular architecture (data, features, labeling, models, ranking, backtest, live-data, dashboard) powering real-time stock ranking predictions.
- Integrated Zerodha Kite Connect for live market data with a yfinance fallback, adding reliability handling for upstream data-source failures.
- Validated system correctness with 29 automated pytest tests, catching 3 production bugs through code review before deployment.
- Deployed an interactive Streamlit dashboard with Plotly candlestick charts, turning model output into a usable, real-time application.

Binance Futures Testnet Trading Bot — AI Trading Backend & API Engineering

2026

Python, REST APIs, OOP, Dependency Injection, Unit Testing

- Built a modular Python trading backend on REST APIs using dependency injection and OOP for testable, swappable components.
- Added input validation, structured logging and exponential-backoff retry logic to make automated order execution resilient to transient API failures.
- Diagnosed and fixed a python-binance library bug blocking Futures Testnet order execution; verified the fix with 34 unit tests.

Cyberbullying Detection — AI Text-Moderation Application

2025

Python, NLP, TF-IDF, Flask, AI Inference

- Built and deployed an AI text-classification application (TF-IDF + tokenization + stopword removal) that detects toxic/abusive text across multiple categories through a live Flask web app.
- Tuned the model for imbalanced data with class-weighted training to improve minority-class detection.
- Served the model through a Flask inference endpoint and paired it with an integrated emotional-support response layer for affected users — connecting AI inference directly to a deployed, user-facing application.

PUBLICATION & CERTIFICATION

- “Jarvis — AI-Powered Voice Assistant,” IJARESM, 2023 — Python voice assistant automating 15+ commands via SpeechRecognition and SMTP.
- Google IT Support Professional Certificate — Coursera / Google.

ACHIEVEMENTS

- Shipped 3+ end-to-end AI/ML projects spanning data engineering, model training, testing and deployment, backed by 63+ automated tests (29 pytest + 34 unit tests).
- Built and shipped QuantEdge across 10 development phases in 7 days, from data pipeline to deployed dashboard.